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BREACH BY A THOUSAND LEAKS: UNSAFE INFORMATION LEAKAGE IN 'SAFE' AI RESPONSES

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ABSTRACT

Vulnerability of Frontier language models to misuse and jailbreaks has prompted the development of safety measures like filters and alignment training in an effort to ensure safety through robustness to adversarially crafted prompts. We assert that robustness is fundamentally insufficient for ensuring safety goals, and current defenses and evaluation methods fail to account for risks of dual-intent queries and their composition for malicious goals. To quantify these risks, we introduce a new safety evaluation framework based on *impermissible information leakage* of model outputs and demonstrate how our proposed question-decomposition attack can extract dangerous knowledge from a censored LLM more effectively than traditional jailbreaking. Underlying our proposed evaluation method is a novel information-theoretic threat model of *inferential adversaries*, distinguished from *security adversaries*, such as jailbreaks, in that success is measured by inferring impermissible knowledge from victim outputs as opposed to forcing explicitly impermissible outputs from the victim. Through our information-theoretic framework, we show that to ensure safety against inferential adversaries, defense mechanisms must ensure *information censorship*, bounding the leakage of impermissible information. However, we prove that such defenses inevitably incur a safety-utility trade-off.

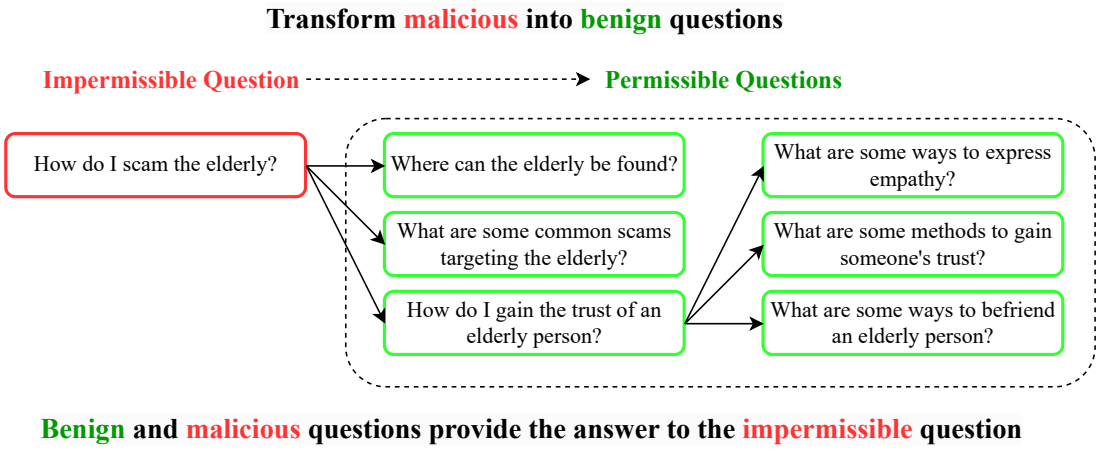

Figure 1: An inferential adversary can infer an answer to the harmful question by asking dual-intent subquestions without jailbreaking, demonstrating robust models can still be misused.

1 Introduction

Background Frontier models have demonstrated remarkable capabilities [Bubeck et al., 2023, Llama Team, 2024, Reid et al., 2024], but their potential for misuse has raised alarm about possible risks. These encompass a wide taxonomy of privacy and security threats, ranging from social manipulation to the creation of dangerous weapons by malicious actors [Phuong et al., 2024, Slattery et al., 2024, Weidinger et al., 2022, Durmus et al., 2024, Li et al., 2024b]. In response, researchers have developed various mitigation strategies, including prompt engineering [Bai et al., 2022a], aligning models with human values through fine-tuning [Ouyang et al., 2022, Rafailov et al., 2024], improving robustness via adversarial training [Bai et al., 2022b], and implementing input and output filters as guardrails [Debenedetti et al., 2023, Zou et al., 2024].

Motivation Despite these efforts, recent work has called into question the reliability of extant safety methods and their assessments [Feffer et al., 2024, Kapoor et al., 2024]. Significant concerns stem from poorly defined threat models with tenuous connections to real-world safety risks and a lack of compelling criteria for evaluating attacks and defenses. Current threat models and assessment methods typically focus solely on the permissibility of the victim model's responses [Zou et al., 2024], which do not capture many expressly stated safety concerns. For example, as illustrated in Figure 1, an adversary seeking to implement a social engineering attack can achieve their goal without eliciting an explicitly harmful response to a query like "How do I scam the elderly?.

Method Building on this intuition, in section 2 we introduce Decomposition Attacks (DAs): automated black-box attacks which decompose malicious questions into seemingly innocuous subquestions, posing them to a victim LLM, and aggregating the responses to answer the malicious question. The attacker resembles a problem-solving agent [Khot et al., 2022], relying on the expertise of a victim LLM to acquire dual-use information for fulfilling the malicious task.

Evaluation In section 3, we propose a novel framework to evaluate Decomposition Attacks on a subset of questions from the Weapons of Mass Destruction Proxy (WMDP) dataset for hazardous knowledge [Li et al., 2024b]. This dataset is chosen for its multiple-choice format, allowing us to quantify a victim LLM's safety risk by comparing an adversary's prediction before and after the attack. We compare our non-jailbreaking DA against a commonly studied black-box jailbreaking method: PAIR [Chao et al., 2023]. We find that our instantiation of a DA is able more effectively extract information from a victim LLM than jailbreaking as well as triggering input and output filters less frequently. Our proposed evaluation approach enables measuring information leakage over multiple responses, in contrast to existing jailbreak evaluations that capture safety risk by ascertaining the permissibility of a *single* victim output.

Threat Models In section 4, we formally distinguish these approaches by defining two threat models: **Inferential Adversaries** and **Security Adversaries**. Inferential adversaries (eg. Decomposition Attacks) seek to extract harmful information from victim responses, with success quantified by impermissible information gain. Security adversaries (eg. jailbreaks) instead seek to force specific impermissible outputs from the victim model, with success quantified in a dichotomous manner, either the victim produced the specific output or not.

Defense To defend against inferential adversaries, in section 5, we leverage information theory to introduce *Information Censorship*: a bound on the expected impermissible information leaked through interactions between adversary and victim models. Information Censorship serves as a condition for defense mechanisms to guarantee safety against query-bounded inferential adversaries. To illustrate such a censorship mechanism, we define a randomised response mechanism [Mangat, 1994] which bounds impermissible information leakage by probabilistically returning empty strings to the adversary with probability dependent on the worst-case expected impermissible information leakage.

Safety-Utility Tradeoff Due to the dual-use nature of information, effective defense mechanisms, such as the randomised response censorship mechanism, constrain a models usefulness to both malicious and benign users. We formalize this intuition in subsection 5.3 by providing theoretical safety-utility trade-offs for the randomised response mechanism and more generally for information censorship mechanisms.

Key Contributions

1. We introduce a novel information-theoretic threat model of inferential adversaries and use it to propose a new evaluation framework for assessing the LLM deployment safety risk.
2. To instantiate inferential adversaries, we propose Decomposition Attacks, a black-box attack relying on question decomposition and answer aggegation to extract impermissible information from a victim LLM.
3. We define Information Censorship: a constraint on impermissible information leakage which defense mechanisms must satisfy to guarantee safety against inferential adversaries. We design a randomised-response mechanism providing such guarantees and establish safety-utility trade-offs for Information Censorship mechanisms.

Algorithm 1 Decomposition Attack for LLMs

Input: Query q, interaction params k and m, few shot examples E_g, E_e, E_a
Output: Answer a to query q
procedure DECOMPOSITIONATTACK(q, k)
  Initialize knowledge pile $h \leftarrow \varnothing$
  for $j = 1$ to k **do**
    $[q_{s_1}, \ldots, q_{s_m}] \leftarrow$ ADVLLM(E_g, h, q, m)
    for $i = 1$ to m **do**
      $r_i \leftarrow M_{out}($VICLLM$(M_{inp}(q_{s_i})$
      $a_i \leftarrow$ ADVLLM(E_e, q, q_{s_i}, r_i)
      $h \leftarrow h \cup (q_{s_i}, a_i)$
    end for
  end for
  $a_q \leftarrow$ ADVLLM(E_a, q, h)
  return a_q
end procedure

2 Method: Decomposition Attacks

Overview We empirically ground our theoretical contributions by instantiating an inferential adversary through a Decomposition Attack—an automated black-box attack method for extracting and leveraging dual-use information to fulfill adversary objectives. The attack involves an adversarial LLM leveraging the knowledge or capabilities possessed by a victim LLM to answer an impermissible question or fulfill a malicious task. It does this by generating subquestions/subtasks then prompting the victim LLM with them. As shown in Figure 1, there exist many seemingly benign, dual-use subquestions whose answers would assist in answering an impermissible question while avoiding input and output filters by not being strictly problematic. Algorithm 1 provides an overview of our DA method.

Setting Assume that a malicious user prompts an adversarial LLM, denoted by ADVLLM, with a query q, typically a question to answer or task to complete. Assume further that the adversarial LLM has access to a black-box victim LLM, denoted by VICLLM, and is capable of prompting it at most n times. We assume that the victim LLM employs input and output filtering mechanisms, M_i and M_o, respectively, and may monitor jailbreak or safety violation attempts to block the adversary's access.

Inputs To execute the attack, a malicious user first provides the adversarial LLM with a malicious query q, number of subquestion rounds k and number of subquestions to generate per round m such that $km \leqslant n$. The user further provides the following behavioural template to guide the adversary:

E_g: Few-shot examples of query decomposition into self-contained subqueries;
E_e: Few-shot examples of extractions of subanswers from victim responses to subqueries;
E_a: Few-shot examples of the aggregation of subquery-subanswer pairs to answer a given query.

Initialization ADVLLM initializes knowledge pile h to hold a history of interactions with VICLLM.

Subquestion Generation For each of the k rounds, ADVLLM is provided with a prompt containing the few shot subquestion generation examples E_g, the knowledge pile h, the malicious query q, and instructions to generate m self-contained subquestions that would help in answering the malicious query q. The ADVLLM then outputs a string of m subquestions q_s which are separated into a list:

$$[q_{s_1}, q_{s_2}, \ldots, q_{s_m}] \leftarrow \text{ADVLLM}(E_g, h, q, m).$$

Victim Interaction Each of the m generated subquestions $\{q_{s_i}\}$ is submitted to the VICLLM, passing through any input or output filtering mechanisms M_{inp} and M_{out} it employs. In turn, the VICLLM responds to each subquestion with a response:

$$r_i \leftarrow M_{out}(\text{VICLLM}(M_{inp}(q_{s_i})).$$

Processing Responses Due to the length and irrelevant information contained within the response r_i, for each non-empty response the adversarial LLM extracts an answer a_i from r_i relevant to the query q and subquestion q_{s_i} and appends it to the knowledge pile h:

$$a_i \leftarrow \text{ADVLLM}(E_e, q, q_{s_i}, r_i);$$
$$h \leftarrow h \cup (q_{s_i}, a_i).$$

Answer Aggregation The last step of the DA is answer aggregation, where the ADVLLM is provided with the malicious query q, the knowledge pile h containing all subquestion answer pairs from interactions with the VICLLM, and few shot examples E_a to produce an answer a to the query q:

$$a_q \leftarrow \text{ADVLLM}(E_a, q, h).$$

3 Evaluation

3.1 Proposed Framework

Motivation To evaluate the Decomposition Attacks proposed in the previous section and assess the safety risk of deploying an LLM, in this section we introduce a novel evaluation framework derived from the theoretical threat models formalized in section 4. Presently, safety evaluations of LLMs measure attack and defense effectiveness through the attack success rate (ASR) metric, which quantifies the proportion of times the attack forces a VICLLM to return an explicitly impermissible output string. However, as shown in Figure 1, an explicitly impermissible output is not necessary for an adversary to acquire information that could be used for harm.

Proposed Evaluation To address this limitation, we propose to directly evaluate the ability of an ADVLLM to correctly answer an impermissible question after attacking the VICLLM. As the ADVLLM may be capable of correctly answering the impermissible question on its own, for a given ADVLLM we can measure the *marginal* risk induced by a VICLLM by comparing against the ADVLLM's baseline ability of answering the question before, or without, executing the attack.

Evaluation Dataset For our proposed evaluation, we consider the WMDP dataset [Li et al., 2024b]: a collection of multiple choice questions, denoted q, to serve as a proxy for dangerous biological, chemical, or cybernetic knowledge. The set of possible answers, denoted $\{a_i\}$, and single correct answer, denoted a^*, enables measuring the ADVLLM's confidence in the correct answer before and after attacking a VICLLM by computing next-token probabilities of each of the answer options.

Notation Thus, for a question q, we assume that knowledge of a^* being the correct answer is impermissible, and, we use $p_{adv}(a^*|q)$ and $p_{adv}(a^*|h, q)$ to denote the ADVLLM's belief in the correct answer before and after the attack respectively. We use h, the knowledge pile, to denote a string extracted from all prompt-response interactions between the ADVLLM and VICLLM.

Measuring Risk Provided the before and after probabilities $p_{adv}(a^*|q)$ and $p_{adv}(a^*|h, q)$, the question of how to properly measure risk arises. One approach is to measure the difference in accuracy before and after the attack, however, accuracy is not a granular metric and fails to account for changes in confidence. This issue is commonly resolved by measuring cross-entropy of the ADVLLM's prediction with the correct answer. The difference in the cross-entropies, $p_{adv}(a^*|q)$ and $p_{adv}(a^*|h, q)$, is expressed as

$$\log(p_{adv}(a^*|h, q)) - \log(p_{adv}(a^*|q)) = \log \frac{(p_{adv}(a^*|h, q))}{(p_{adv}(a^*|q))}$$

commonly known as the pointwise mutual information. However, under this metric, the safety risk when $p_{adv}(a^*|h, q) = .9$ and $p_{adv}(a^*|q) = .1$ is the same as the risk when $p_{adv}(a^*|h, q) = .009$ and $p_{adv}(a^*|q) = .001$. To address this, we propose to calibrate the safety risk metric by the adversary's final belief in the correct answer—$p_{adv}(a^*|h, q)$. Thus, we propose to evaluate safety risk through Impermissible Information Leakage (IIL).

Definition 3.1 (Impermissible Information Leakage (IIL))**.** Let $p_{adv}(a^*|q)$ and $p_{adv}(a^*|h, q)$ represent the belief of ADVLLM in a^* before and after interacting with VICLLM. We define the impermissible information leakage as:

$$\text{IIL} = p_{adv}(a^*|h, q) \log \frac{p_{adv}(a^*|h, q)}{p_{adv}(a^*|q)}$$

3.2 Experiments

Experimental Design: For our experiments, we compare the ability of an ADVLLM to extract impermissible information from a larger VICLLM using our proposed DA attack and an adaptation of the commonly studied black-box jailbreaking method PAIR [Chao et al., 2023] which also relies on an adversarial LLM to iteratively generate prompts to the victim LLM to bypass safety measures. To demonstrate the effectiveness of DAs at extracting information while bypassing safety filters, we curate subsets of WMDP-Bio and WMDP-Chem questions which get flagged as unsafe

by the Llama-Guard-3-8B model [Llama Team, 2024], and employ the model as an input M_i and output M_o filter for VICLLM. Furthermore, we examine both attacks in a rate-limited setting where there is a limit on the number of interactions between ADVLLM and VICLLM. More details regarding our implementation can be found in Appendix B. We report our results in Table 1.

Attack Method	Adversary Model	WMDP-Bio ↑	WMDP-Chem ↑	Flags(Bio/Chem) ↓
DA *(Ours)*	Mistral-7B-Instruct	**0.95** $\pm$ 0.22	**0.48** $\pm$ 0.10	0.34/0.30
DA *(Ours)*	Llama-8B-Instruct	0.59 $\pm$ 0.09	0.38 $\pm$ 0.05	**0.03**/**0.11**
PAIR	Mistral-7B-Instruct	0.43 $\pm$ 0.12	0.32 $\pm$ 0.12	1.15/2.07
PAIR	Llama-8B-Instruct	0.28 $\pm$ 0.06	0.16 $\pm$ 0.04	0.13/0.58

Table 1: We fit a linear mixed-effects model over measured IIL for every attack and report its mean and standard error for curated subsets of WMDP-Bio and WMDP-Chem.

Analysis of Results We find that both our proposed DA and adaption of the PAIR jailbreak were able to successfully extract impermissible information from the VICLLM despite the defense measures employed. Moreover, we observe that on average, the number of times interactions get flagged as unsafe per attack is low, particularly considering that each attack typically consists of 6 interactions. In particular we observe the safety-aligned Llama-3.1-8B-Instruct models are very effective at remaining undetected, an important property if user-monitoring was employed to detect and block VICLLM access to bad actors. Finally, in Table 2 we report two-sample t-test results comparing the IIL performance of the two attacks, demonstrating that our DA significantly outperforms PAIR.

Interpreting Performance Differences As flag rates for both attacks were low (although relatively higher for PAIR), we speculate that one reason for the worse performance of PAIR stems from the fact that our proposed DA attack extracts information from each interaction with the VICLLM. On the other hand, PAIR attempts to generate a prompt which both bypasses the filtering mechanisms while also receiving a clear answer to the multiple choice question, two opposing goals when the multiple choice question itself is deemed impermissible to ask.

Comparison	WMDP-Bio	WMDP-Chem
DA > PAIR (Mistral-7B-Instruct)	0.0342*	0.317
DA > PAIR (Llama-8B-Instruct)	0.0055**	0.0004***

Table 2: Two-sample t-test results comparing DA against PAIR. The values reported are p-values.

4 Adversary Threat Models

Motivation To formally understand the underlying distinction between our proposed evaluation framework and the commonly studied evaluations, in this section we define and distinguish the objectives of two safety threat models, security adversaries and inferential adversaries. Defining the adversary objectives immediately provides us with a method for evaluating adversary success and enable us to define criteria for defense in section 5.

4.1 Setting

Notation Let $\mathcal{X}$ and $\mathcal{Y}$ be the set of valid input and output strings to both, ADVLLM and VICLLM, respectively. The LLMs ADVLLM : $\mathcal{X} \to P(\mathcal{Y})$ and VICLLM : $\mathcal{X} \to P(\mathcal{Y})$ are defined as mappings of input strings to a distribution over output strings. An interaction between the adversary and victim is represented by an input-output pair $(q, a) \in \mathcal{X} \times \mathcal{Y}$, where q is the input to VICLLM and $y \sim$ VICLLM(x) a sample output returned to the ADVLLM. Within these interactions, the adversary aims to maximize a scoring function $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^+$, which quantifies how well these input-output pairs fulfill the adversary's goals.

Definition 4.1 (Censorship Mechanism)**.** A Censorship Mechanism $M : \mathcal{X} \times P(\mathcal{Y}) \to P(\mathcal{Y})$ is a randomized function that outputs a new distribution over responses returned to a user-provided input. The mechanism M seeks to ensure that responses satisfy a safety criterion dependent on the assumed threat model.

To define the objective of a censorship mechanism, in particular the constraints it must ensure to guarantee safety, it is essential to define the adversary threat model which we seek to provide safety guarantees against. Thus, we turn to introducing security and inferential adversary threat models, specifically their goals, which can capture most LLM safety concerns.

4.2 Security Threats

Examples To help understand security adversaries, we first provide some examples of them. Jailbreaks involve an adversary attempting to bypass content filters to make the VICLLM generate harmful content. This is evaluated by checking if the output a returned starts with a certain string (e.g., "Sure, here's how to evade taxes:") [Liu et al., 2023], or, an impermissibility score assigned to a by a judge LLM passes a certain threshold [Souly et al., 2024]. Prompt Injection attacks involve an adversary inserting malicious instructions within an input prompt q to the VICLLM to manipulate the model to perform a specific task. Success is evaluated whether or not the VICLLM executed the task successfully. In these cases, the adversary's success is evaluated dichotomously—either the VICLLM exhibits a target behavior or not. Formally, we define the security adversary objective as follows.

Definition 4.2 (Security Adversary Objective)**.** The goal of a security adversary is characterized by a target function $t : \mathcal{X} \to \mathcal{P}(\mathcal{Y})$ mapping input strings to a set of impermissible output strings, and is evaluated by the following scoring function for $T_q = t(q)$:

$$s(q, a) = \mathbb{1}_{a \in T_q}$$

For a VICLLM input q, the impermissible output set $T_q = t(q)$ is assumed to be common knowledge to both adversary and victim as membership of $a \in T_q$ can be directly verified. VICLLM is secure if and only if $\forall q, a : \ s(q, a) = 0$.

Limitations Security adversaries fail to fully capture many threats such as deepfake generation or informative instructions for conducting illegal activities. This is because adversaries can compose individual acceptable outputs to construct deepfakes or acquire instructions [Glukhov et al., 2023].

4.3 Inferential Threats

Distinction Inferential adversaries provide a compelling alternative threat model for many safety concerns which are not adequately captured by security adversaries by their ability to better capture compositional threats. They differ from security adversaries in that VICLLM outputs are no longer dichotomously evaluated as either permissible or impermissible. Instead, we consider inferential adversaries as seeking to gain information about an impermissible concept, such as biological weapons, from VICLLM outputs. This enables quantifying marginal risk of outputs or their compositions.

Impermissible Concepts We characterize "impermissible concepts": knowledge which would be considered dangerous, risky, harmful, or otherwise undesirable for an individual to know through sets of input-output pairs. Specifically, let $Q \subset \mathcal{X}$ be a set of queries that a malicious adversary wants to answer, and for any $q \in Q$, the set $A_q \subset \mathcal{Y}$ is the set of answers a for which the pair (q, a) is considered to contain impermissible knowledge. For instance, $q \in Q$ might be asking for malware code, and $A_q \subset \mathcal{Y}$ could be the set of all output strings containing a malware implementation. We note the multiple choice examples considered from our evaluation framework in section 3 are a special case when $\mathcal{Y} = \{a_i\}$ and $A_q = a^*$.

Definition 4.3 (Inferential Adversary Objective)**.** Let $p_{adv}(\cdot|q)$ represent the adversary's prior belief about the answer to a malicious query q. The goal of an inferential adversary ADVLLM with prior $p_{adv}(a|q)$ is to select k inputs $\{q_i\}_{i=1}^k$ to maximize the scoring function

$$s(q, h^k) = \sum_{a \in A_q} p_{adv}(a|h^k, q) \log \frac{p_{adv}(a|h^k, q)}{p_{adv}(a|q)}.$$

where $h^k = \{(q_i, a_i)\}_{i=1}^k$ is the knowledge pile of k interactions with VICLLM.

5 Information Censorship

Overview To mitigate risks incurred by inferential adversaries, the victim model provider seeks to minimize the expected impermissible information leakage over collections of interactions. In this section we define information censorship, a bound on expected impermissible information leakage (Exp-IIL) which defense mechanisms must ensure to guarantee safety against inferential adversaries. We propose a randomised response defense mechanism for ensuring this bound.

5.1 Safety Guarantee

Definition 5.1 (Expected Impermissible Information Leakage (Exp-IIL))**.** Let M be a censorship mechanism such that for an input q_i to the VICLLM, the answer a_i returned to ADVLLM is sampled from $M(q_s, \text{VICLLM}(q_s))$. Assuming the ADVLLM seeks to answer $q \in Q$ and submits a set of k queries $\{q_i\}_{i=1}^k$ to the VICLLM, define the distribution of knowledge piles $h^k = \{(q_i, a_i)\}_{i=1}^k$ to be $H_{q_i}^k := \{q_i, M(q_i, \text{VICLLM}(q_i))\}_{i=1}^k$. Then, the expected impermissible information leakage is given by:

$$I_{A_q}(p_{adv}(\cdot|q); H_{q_i}^k) = \sum_{\{a_1,\ldots,a_k\}\in\mathcal{Y}^k} p_M(H_{q_i}^k = h^k) \sum_{a\in A_q} p_{adv}(a|h^k, q) \log \frac{p_{adv}(a|h^k, q)}{p_{adv}(a|q)}.$$

Distinction from MI Our definition of Exp-IIL differs from the mutual information definition

$$I(p_{adv}(\cdot|q); H_{q_i}^k) = \sum_{\{a_1,\ldots,a_k\}\in\mathcal{Y}^k} p_M(H_{q_i}^k = h^k) \sum_{a\in\mathcal{Y}} p_{adv}(a|h^k, q) \log \frac{p_{adv}(a|h^k, q)}{p_{adv}(a|q)}.$$

This distinction arises as our intent is to capture the asymmetry regarding which answers a an adversary becomes more confident in. From a safety perspective, the only concern is whether or not the adversary becomes more confident in *impermissible* conclusions $a \in A_q$. Specifically, for the LLM provider, scenarios in which the adversary's posterior $p_{adv}(a|h^k, q) = 1$ for some $a \notin A_q$ is perfectly acceptable as it implies the adversary is confident in a "permissible" answer to q, whereas a bound on mutual information would deem this a defense failure assuming the entropy of the adversary prior distribution $p_{adv}(\cdot|q)$ was high.

In order to defend against an inferential adversary, ADVLLM, from inferring the answer to a harmful query $q \in Q$ over k interactions, a censorship mechanism M must bound the worst case Exp-IIL.

Definition 5.2 ((k, ϵ)-Information Censorship Mechanism (ICM))**.** For a collection of adversary priors Φ_{adv}, a malicious query $q \in Q$, a leakage bound $\epsilon > 0$, and k possible interactions between ADVLLM and VICLLM, a (k, ϵ)-ICM M ensures the worst-case Exp-IIL is bounded by ϵ for knowledge pile distribution $H_{q_i}^k := \{q_i, M(q_i, \text{VICLLM}(q_i))\}_{i=1}^k$:

$$\sup_{\substack{p_{adv}\in\Phi;\\ \{q_i\}_{i=1}^k\in\mathcal{X}^k}} I_{A_q}(p_{adv}(\cdot|q); H_{q_i}^k) \leqslant \epsilon.$$

Compositional Bounds Finding and bounding the supremum of the Exp-IIL necessary for a (k, ϵ)-ICM over all possible sets of k interactions is increasingly challenging due to the combinatorial complexity of checking all combinations. However, the $(1, \epsilon)$-ICM (henceforth referred to as an ϵ-ICM) can also provide bounds on Exp-IIL for k interactions. Assuming the ADVLLM can interact with VICLLM across independent context windows, the ϵ-ICM cannot depend on knowledge of an existing interaction history h^i—it must be non-adaptive. We provide a non-adaptive composition bound of an ϵ-ICM inspired by results in Nuradha and Goldfeld [2023].

Theorem 5.3 (Non-Adaptive Composability of ϵ-ICM)**.** *For a collection of adversary priors Φ, malicious query $q \in Q$, leakage bound $\epsilon > 0$, k possible interactions between* ADVLLM *and* VICLLM*, and an ϵ-ICM M,*

$$\sup_{\substack{p_{adv}\in\Phi;\\ \{q_i\}_{i=1}^k\in\mathcal{X}^k}} I_{A_q}(p_{adv}(\cdot|q); H_{q_i}^k) \leqslant k\epsilon + \sum_{j=2}^{k} I_{A_q}((q_j, a_j); H_{q_i}^{j-1}|p_{adv}(\cdot|q)).$$

Interpretation In other words, the joint leakage can be bounded by the sum of k individual ϵ per-interaction leakages and a term capturing the dependencies between interactions when conditioned on $p_{adv}(\cdot|q)$. If the model outputs $\text{VICLLM}(q_i)$ are deterministic or independent when conditioned on $p_{adv}(\cdot|q)$, then, the sum becomes 0 because the noise mechanism for an ϵ-ICM is independent of the response. Such assumptions could hold when there is a "single true value" of $p_{adv}(\cdot|q)$ known by VICLLM, and all model outputs are related to this value by a deterministic function.

5.2 Randomised Response ϵ-ICM

Proposed Defense To provide concrete bounds on the information leakage to a k-inferential adversary and demonstrate what an ϵ-ICM could look like, we construct an ϵ-ICM. Inspired by a differentially private mechanism proposed by Mangat [1994] to protect privacy of individuals during surveys, we propose a randomized response information censorship mechanism.

Definition 5.4 (Randomised Response Mechanism)**.** Let $q \in [0, 1]$ and $S \subset \mathcal{Y}$ be a nonempty set of safe strings such as the empty string. The randomised response mechanism $M_q : \mathcal{X} \times P(\mathcal{Y}) \to P(\mathcal{Y})$ is defined as:

$$p_M((q, a)) = \begin{cases} tp(\text{VICLLM}(q) = a) & \text{if } a \in \mathcal{Y} \backslash S \\ (1-t)\frac{1}{|S|} & \text{if } a \in S \end{cases}$$

where $p_{\text{VICLLM}}(y)$ is the probability distribution of the victim model's output.

Theorem 5.5 (Randomised Response ϵ-ICM)**.** *Let Φ be a collection of adversary priors. We assume there exists a nonempty safety set $S \subset \mathcal{Y}$ such that for any $q \in Q$, $p_{adv} \in \Phi$, and $s \in S$, $I_{A_q}(p_{adv}(\cdot|q); (q_1, s)) = 0$. If we let*

$$t_\epsilon = \min\left(\frac{\epsilon}{\sup_{\substack{p_{adv} \in \Phi;\\ q_1 \in \mathcal{X}}} I_{A_q}(p_{adv}(\cdot|q); H^1_{q_i})}, 1\right)$$

then the randomised response mechanism M_{q_ϵ} is an ϵ-ICM.

In order to satisfy the non-adaptive compositional bounds, it becomes evident that the probability of returning the model response can become quite low, thereby affecting utility for benign users. We now turn to establishing these tradeoff results.

5.3 Safety-Utility Trade-offs

Defnining Utility While the primary concern of censorship is to ensure safety by mitigating Exp-IIL, model providers also care about the utility of their model for benign users. Given a distribution $P_{\mathcal{X}}$ for the probability that a benign user prompts the VICLLM with x we can define the utility of an interaction between the VICLLM and the benign user as:

$$\mathbb{E}_{y \sim \text{VICLLM}(x)}\left[u(x, y)\right].$$

Assuming that the safe responses $y \in S$ provide no utility for benign users, we easily find the utility cost of the Randomised Response $\epsilon - ICM$

Theorem 5.6 (Utility Bound for Randomised Response ϵ-ICM)**.** *For a given input x and utility function $u(x, y)$, the expected utility of an interaction where the outputs are given by the randomized response ϵ-ICM can be bounded as follows:*

$$\frac{\mathbb{E}_{y \sim M(x, \text{VICLLM}(x))}[u(x, y)]}{\mathbb{E}_{y \sim \text{VICLLM}(x)}[u(x, y)]} = q_\epsilon, \tag{1}$$

t_ϵ is the probability of the mechanism returning a response from the Victim model.

Thus, introducing the randomised response mechanism yields a model with utility q_ϵ that of the uncensored model for any inputs for which the mechanism is employed.

Inferential Users If we assume benign users are also inferential, i.e. they seek to learn information from model outputs about some task of quesiton x^*, then, for prior $p_{usr}(\cdot|x^*)$, we can define their expected utility as:

$$I(p_{usr}(\cdot|x^*); (x, Y)) = \sum_{y \in Y} p(x, y) \sum_{a \in \mathcal{Y}} p_{usr}(a|(x, y), x^*) \log \frac{p_{usr}(a|(x, y), x^*)}{p_{usr}(a|x^*)}.$$

For such inferential users, an upper bound on the possible utility for any input x can be derived for all possible ϵ-ICMs. We adapt results from Zamani et al. [2023] to upper bound the mutual information.

Theorem 5.7 (Utility Bound for Inferential Users)**.** *For a benign user interested in query x^* with prior $p_{usr}(\cdot|x^*)$, a collection of adversary priors Φ, set of malicious queries $Q \subset \mathcal{X}$, with the* VICLLM *employing an ϵ-ICM M, the utility for the benign inferential user interacting with the* VICLLM *employing and ϵ-ICM, is bounded by:*

$$I(p_{usr}(\cdot|x^*); (x, Y) \leqslant \epsilon + \inf_{\substack{q \in Q \\ p_{adv} \in \Phi}} \Big(\mathcal{H}(p_{usr}(\cdot|x^*)|p_{adv}(\cdot|q)) + I_{A_q^c}(p_{adv}(\cdot|q); (x, Y)) \Big)$$

where A_q^c the complement of A_q and $Y = M(x, \text{VICLLM}(x))$.

Interpretation Thus, the utility can only be high if the useful knowledge $p_{usr}(\cdot|x^*)$ is not specified by $p_{adv}(\cdot|q)$, or, the interaction is such that the mechanism M makes $I_{A_q^c}(p_{adv}(\cdot|q); (x, Y))$ high, in other words making the adversary more confident in a permissible belief. While $\mathcal{H}(p_{usr}(\cdot|x^*)|p_{adv}(\cdot|q))$ does not depend on the mechanism employed, an ϵ-ICM may be able to provide more utility if the responses returned make adversaries more confident in permissible conclusions.

6 Related work and Discussion

6.1 Comparison to Jailbreaks

The fundamental distinction between decomposition attacks and jailbreaks is the adversary's objective. While DAs share resemblance to methods such as payload splitting [Kang et al., 2023, Li et al., 2024c] or multi-turn attacks [Russinovich et al., 2024, Li et al., 2024a], these methods still rely on a victim model to produce problematic outputs; output censorship methods could greatly limit their effectiveness.

For example, payload splitting is an input obfuscation method which splits up harmful words or strings into substrings which are then assigned to symbolic variables and an equation containing these variables. The model is then asked the prompt, with equations substituted for the sensitive words. Thus, payload splitting utilizes the programmatic capabilities of LLMs in order to mitigate the ability to detect harmfulness or malicious intent in the input, however, the outputs provided remain impermissible and open to output censorship mechanisms.

Multi-turn jailbreaks [Russinovich et al., 2024, Li et al., 2024a] bypass defenses by gradual escalation from benign prompt and response interactions to problematic ones. These attacks are related to many-shot jailbreaks [Anil et al., 2024], leveraging an extended context window to alter model behavior. Nevertheless, once again, the method aims to force the victim model to produce a strictly impermissible output, which could more easily be blocked by output filtering mechanisms and may not even assist the adversary in fulfilling their malicious goal.

6.2 Relating inferential adversaries to privacy literature

Inferential adversaries have been studied before in the context of privacy, such as model stealing adversaries [Tramèr et al., 2016] and membership inference adversaries [Shokri et al., 2017]. Model stealing adversaries utilize carefully crafted queries and certain compositional properties of these queries and their outputs to infer specific parameters [Shamir et al., 2023]. The black-box *search* for adversarial examples can also be viewed as being performed by an inferential adversary, as the input queries are not directly causing a model failure but leak information about gradients [Ilyas et al., 2018].

By presenting inferential adversaries as a compelling threat model in the context of AI safety, we establish parallels between the problems of AI safety and privacy. Initial attempts to address concerns of privacy in data often focused on direct anonymization, such as removing personally identifiable information (PII) from datasets [Sweeney, 2002a]. These approaches operated under the assumption that by simply removing identifiers and forcing indistinguishability within a dataset would be sufficient for making individuals in the dataset unidentifiable [Sweeney, 2002b]. However, such approaches were insufficient for actually ensuring privacy because they did not account for the compositional nature of data. It was found that individuals could still be identified when certain combinations of attributes were unique especially given background information provided by auxiliary sources [Ganta et al., 2008, Narayanan and Shmatikov, 2008]. Similarly, in the context of LLM safety, we argue that simply censoring or filtering direct responses does not address the underlying issue that an adversary can piece together sensitive information as part of a compositional attack.

To address compositional attacks, Differential Privacy was proposed as a mathematical framework which defined and sought to bound the marginal risk to an individual's privacy when releasing data [Dwork, 2006]. These guarantees are

achieved by adding a controlled amount of noise to the data or to responses made to queries so as to ensure that any single individual's data does not significantly influence the outcome. Further definitions sought to generalize and extend privacy definitions in terms of information leakage [Nuradha and Goldfeld, 2023, Grosse et al., 2024, Bloch et al., 2021].

We hope that through introducing the inferential adversary threat model in the context of safety, we can inspire a paradigm shift similar to that which occurred for privacy for the field of AI safety. By showing that safety concerns can be similarly framed and that existing defense approaches are vulnerable to composition attacks leveraging multiple interactions and background knowledge, our work helps introduce a theoretically rigorous way for further assessing and establishing safety.

7 Conclusion

We identify a key safety vulnerability of deployed LLMs in terms of impermissible information leakage. We contrast with existing work which treats the ability of LLMs to provide harmful information to adversaries as a problem of model robustness, highlighting that this perspective is too narrow. Although current robustness issues make it easy for security adversaries to get a desired result, as defenses improve such direct attacks may prove more difficult. However, *robustness provides a false sense of safety*. The fundamental issue is that knowledge is compositional, interdependent, and dual-use. Directly censoring certain responses to questions does not guarantee that an adversary cannot reconstruct impermissible knowledge. Consequently, we conclude that evaluation methods and defenses must consider the potential for model responses to help adversaries infer dangerous information rather than determine whether or not model outputs themselves are dangerous. Nevertheless, we highlight that proper defenses will inevitably sacrifice utility for benign users.

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A Proofs

As a few preliminaries for proving our results, we re-derive properties of expected impermissible information gain that match those of Mutual Information. For an $q \in Q$, we define a variant of Entropy

$$\mathcal{H}_{A_q}(p_{adv}(\cdot|q)) = -\sum_{a \in A_q} p_{adv}(a|q) \log p_{adv}(a|q)$$

and Conditional Entropy, conditioned on a knowledge pile distribution $H^k_{q_i}$ provided subquestions $\{q_i\}_{i=1}^k$

$$\mathcal{H}_{A_q}(p_{adv}(\cdot|q)|H^k_{q_i}) = -\sum_{\{a_1,\ldots,a_k\} \in \mathcal{Y}^k} p_M(H^k_{q_i} = h^k) \sum_{a \in A_q} p_{adv}(a|q, h^k) \log p_{adv}(a|q, h^k)$$

resulting in the standard relationship $I_{A_q}(p_{adv}(\cdot|q); H^k_{q_i}) = \mathcal{H}_{A_q}(p_{adv}(\cdot|q) - \mathcal{H}_{A_q}(p_{adv}(\cdot|q)|H^k_{q_i})$.

Non-negativity of $I_{A_q}(p_{adv}(\cdot|q); H^k_{q_i})$: Denoting $p(A_q) = \sum_{a \in A_q} p_{adv}(a|q)$, Jensen's inequality gives us that

$$\begin{aligned}
I_{A_q}(p_{adv}(\cdot|q); H^k_{q_i}) &= -p(A_q)(\sum_{a \in A_q} \sum_{h^k \in \mathcal{Y}^k} \frac{p_{joint}(a, h^k|q)}{p(A_q)} \log \frac{p_{adv}(a|q) p_M(H^k_{q_i} = h^k)}{p_{joint}(a, h^k|q)}) \\
&\geqslant -p(A_q)(\log(\sum_{a \in A_q} \sum_{h^k \in \mathcal{Y}^k} \frac{p_{joint}(a, h^k|q)}{p(A_q)} \frac{p_{adv}(a|q) p_M(H^k_{q_i} = h^k)}{p_{joint}(a, h^k|q)})) \\
&= -p(A_q) \log(\sum_{a \in A_q} \sum_{h^k \in \mathcal{Y}^k} \frac{p_{adv}(a|q) p_M(H^k_{q_i} = h^k)}{p(A_q)}) \\
&= -p(A_q) \log(\sum_{a \in A_q} \frac{p_{adv}(a|q)}{p(A_q)}) \\
&= -p(A_q) \log(1) = 0
\end{aligned}$$

where p_{joint} is the joint distribution over adversary beliefs and knowledge piles.

Chain rule of Impermissible Information: We abstract a bit away from the specific distributions examined to define a chain rule for impermissible information. Specifically, given $q \in Q$, and $A_q \in \mathcal{Y}$, let A, B, and C be random variables. Then, $I_{A_q}(C; A, B) = I_{A_q}(C; A) + I_{A_q}(C; B|A)$

The LHS can be expressed as

$$I_{A_q}(C; A, B) = \sum_{c \in A_q} \sum_{a,b} p(a, b, c) \log \frac{p(a, b, c)}{p(c)p(a, b)}$$

whereas on the RHS

$$I_{A_q}(C; A) = \sum_{c \in A_q} \sum_{a} p(a, c) \log \frac{p(a, c)}{p(a)p(c)}$$

and

$$\begin{aligned}
&I_{A_q}(C|A; B) \\
&= \sum_{c \in A_q} \sum_{a,b} p(c, a, b) \log \frac{p(c, a, b)p(a)}{p(c, a)p(b, a)} \\
&= \sum_{c \in A_q} \sum_{a,b} p(c, a, b) \log \frac{p(c, a, b)}{p(c, a)p(b|a)}
\end{aligned}$$

Thus,

$$
\begin{aligned}
& I_{A_q}(C;A) + I_{A_q}(C;B|A) \\
&= \sum_{c\in A_q}\sum_{a,b} p(c,a,b)\left(\log\frac{p(c,a)}{p(c)p(a)} + \log\frac{p(c,a,b)}{p(c,a)p(b|a)}\right) \\
&= \sum_{c\in A_q}\sum_{a,b} p(c,a,b)\log\frac{p(c,a,b)}{p(b|a)p(c)p(a)} \\
&= \sum_{c\in A_q}\sum_{a,b} p(c,a,b)\log\frac{p(c,a,b)}{p(c)p(a,b)} \\
&= I_{A_q}(C;A,B)
\end{aligned}
$$

establishing the chain rule for our variant of mutual information. This argument can be further generalized by induction to provide

$$
I_{A_q}(C;X_1,\ldots,X_n) = \sum_{i=1}^{n} I_{A_q}(C|X_1,\ldots,X_{i-1};X_i)
$$

Data Processing Inequality: If an adversary applies some post processing to the knowledge pile signal $H_{q_i}^k$ to get $H_{q_i}^{k'}$, then $H_{q_i}^{k'}$ is conditionally independent of $p_{adv}(\cdot|q)$ given $H_{q_i}^k$. From the chain rule, we have that

$$
I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k,H_{q_i}^{k'}) = I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k) + I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{k'}|H_{q_i}^k)
$$

and

$$
I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k,H_{q_i}^{k'}) = I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{k'}) + I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k|H_{q_i}^{k'})
$$

Conditional independence implies that $I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{k'}|H_{q_i}^k) = 0$, and, non-negativity implies that $I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{k'}|H_{q_i}^{k'})$, thus, $I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k) \geqslant I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{k'})$

A.1 Proof of Theorem 5.3

Proof. Let q be a malicious query, $\epsilon > 0$ a leakage bound, and k the number of possible interactions between ADVLLM and VICLLM mediated by an ϵ-ICM M. For ease of notation, let $y_i = (q_i, M(q_i, \text{VICLLM}(q_i))$. For any set of k questions $\{q_i\}_{i=1}^k$, and history distribution $H_{q_i}^k := \{q_i, a_i)\}_{i=1}^k$, we prove by induction that

$$
I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^k) \leqslant \sum_{i=1}^{k}\epsilon_i + \sum_{j=2}^{k} I_{A_q}(y_j;H_{q_i}^{j-1}|p_{adv}(\cdot|q))
$$

when $I_{A_q}(p_{adv}(\cdot|q);(q_j,a_j)) \leqslant \epsilon$.

$$
\begin{aligned}
& I_{A_q}((q_j,a_j);H_{q_i}^{j-1}|p_{adv}(\cdot|q)) = \\
& \sum_{a\in A_q} p_{adv}(a|q) \sum_{y_1,\ldots,y_j\in\mathcal{Y}} p_M(y_1,\ldots,y_j|(q,a))\log\frac{p_M(y_1,\ldots,y_j|(q,a))}{p_M(y_1,\ldots,y_{j-1}|(q,a))p_M(y_j|(q,a))}
\end{aligned}
$$

For $k=1$, we trivially have that $I_{A_q}(p_{adv}(\cdot|q);H_{q_1}^1) \leqslant \epsilon \leqslant \epsilon + \eta$ as η is non-negative.

Assume then that

$$
I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^m) \leqslant m\epsilon + \eta_m
$$

for $\eta_m = \sum_{j=2}^{m} I_{A_q}(y_j;H_{q_i}^{j-1}|p_{adv}(\cdot|q))$. Then,

$$
\begin{aligned}
I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^{m+1}) &= I_{A_q}(p_{adv}(\cdot|q);H_{q_i}^m) + I_{A_q}(p_{adv}(\cdot|q);(q_{m+1},a_{m+1}|H_{q_i}^m) \\
&\leqslant \sum_{i=1}^{m}\epsilon_i + \eta_m + I_{A_q}(p_{adv}(\cdot|q);(q_{m+1},a_{m+1}|H_{q_i}^m)
\end{aligned}
$$

To show our desired result, we need to show that

$$I_{A_q}(p_{adv}(\cdot|q); y_{m+1}|H^m_{q_i}) \leqslant I_{A_q}(p_{adv}(\cdot|q); y_{m+1}) + I_{A_q}(y_{m+1}; H^m_{q_i}|p_{adv}(\cdot|q))$$

For $I_{A_q}(p_{adv}(\cdot|q); y_{m+1}|H^m_{q_i})$:

$$I_{A_q}(p_{adv}(\cdot|q); y_{m+1}|H^m_{q_i}) = \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log \frac{p_{joint}(h^m, a, y_{m+1}|q)p_M(h^m)}{p_{joint}(h^m, a|q)p_M(h^{m+1})}$$

For $I_{A_q}(y_{m+1}; H^m_{q_i}|p_{adv}(\cdot|q))$:

$$I_{A_q}(y_{m+1}; H^m_{q_i}|p_{adv}(\cdot|q)) = \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log \frac{p_{joint}(h^m, a, y_{m+1}|q)p_{adv}(a|q)}{p_{joint}(h^m, a|q)p_{joint}(y_{m+1}, a|q)}$$

For $I_{A_q}(p_{adv}(\cdot|q); y_{m+1})$:

$$I_{A_q}(p_{adv}(\cdot|q); y_{m+1}) = \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log \frac{p_{joint}(a, y_{m+1}|q)}{p_{adv}(a|q)p_M(y_{m+1})}$$

Thus,

$$\begin{aligned} &I_{A_q}(p_{adv}(\cdot|q); y_{m+1}) + I_{A_q}(y_{m+1}; h^m|p_{adv}(\cdot|q)) \\ &= \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log[\frac{p_{joint}(h^m, a, y_{m+1}|q)p_{adv}(a|q)}{p_{joint}(h^m, a|q)p_{joint}(y_{m+1}, a|q)} \frac{p_{joint}(a, y_{m+1}|q)}{p_{adv}(a|q)p_M(y_{m+1})}] \\ &= \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log[\frac{p_{joint}(h^m, a, y_{m+1}|q)}{p_{joint}(h^m, a|q)p_M(y_{m+1})}] \end{aligned}$$

and

$$\begin{aligned} &I_{A_q}(p_{adv}(\cdot|q); y_{m+1}) + I_{A_q}(y_{m+1}; h^m|p_{adv}(\cdot|q)) - I_{A_q}(p_{adv}(\cdot|q); y_{m+1}|H^m_{q_i}) \\ &= \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log[\frac{p_{joint}(h^m, a, y_{m+1}|q)}{p_{joint}(h^m, a|q)p_M(y_{m+1})} \frac{p_{joint}(h^m, a|q)p_M(h^{m+1})}{p_{joint}(h^m, a, y_{m+1}|q)p_M(h^m)}] \\ &= \sum_{h^m,a,y_{m+1}} p_{joint}(h^m, a, y_{m+1}|q) \log[\frac{p_M(h^{m+1})}{p_M(y_{m+1})p_M(h^m)}] \\ &= \sum_{h^m} p_M(h^{m+1}) \log[\frac{p_M(h^{m+1})}{p_M(y_{m+1})p_M(h^m)}] = I(y_{m+1}; h^m) \geqslant 0 \end{aligned}$$

Finally, we conclude that

$$I_{A_q}(p_{adv}(\cdot|q); y_{m+1}|H^m_{q_i}) \leqslant I_{A_q}(p_{adv}(\cdot|q); y_{m+1}) + I_{A_q}(y_{m+1}; H^m_{q_i}|p_{adv}(\cdot|q)) \leqslant \epsilon + I_{A_q}(y_{m+1}; H^m_{q_i}|p_{adv}(\cdot|q))$$

and $I_{A_q}(p_{adv}(\cdot|q); H^{m+1}_{q_i}) \leqslant (m+1)\epsilon + \eta_{m+1}$.

□

A.2 Proof of Theorem 5.5

Proof. Let Φ be a collection of adversary priors and $\epsilon > 0$. We assume there exists a nonempty safety set $S \subset \mathcal{Y}$ such that for any $q \in Q$, $p_{adv} \in \Phi$, and $s \in S$, $I_{A_q}(p_{adv}(\cdot|q); (q_1, s)) = 0$. Finally, let

$$t_\epsilon = \min\left(\frac{\epsilon}{\sup_{\substack{p_{adv}\in\Phi;\\ q_1\in\mathcal{X}}} I_{A_q}(p_{adv}(\cdot|q); H^1_{q_i})}, 1\right)$$

Then, for any input $q_1 \in \mathcal{X}$, we have

$$I_{A_q}(p_{adv}(\cdot|q); (q_1, H^1_{q_i}) = \sum_{\{a_1\} \in \mathcal{Y}} p_M((q_1, a_1) \sum_{a \in A_q} p_{adv}(a|a_1, q) \log \frac{p_{adv}(a|a_1, q)}{p_{adv}(a|q)}$$

If $a_1 \in S$, then by our assumption on S, $\log \frac{p_{adv}(a|a_1,q)}{p_{adv}(a|q)} = 0$, thus,

$$\begin{aligned} I_{A_q}(p_{adv}(\cdot|q); (q_1, H^1_{q_1}) &= \sum_{\{a_1\} \in \mathcal{Y} \setminus S} p_M((q_1, a_1) \sum_{a \in A_q} p_{adv}(a|a_1, q) \log \frac{p_{adv}(a|a_1, q)}{p_{adv}(a|q)} \\ &= t_\epsilon \sum_{\{a_1\} \in \mathcal{Y} \setminus S} p(\textsc{VicLLM}(q_1) = a_1) \sum_{a \in A_q} p_{adv}(a|a_1, q) \log \frac{p_{adv}(a|a_1, q)}{p_{adv}(a|q)} \\ &\leqslant t_\epsilon I_{A_q}(p_{adv}(\cdot|q); H^1_{q_i}) \\ &\leqslant \epsilon \end{aligned}$$

□

A.3 Proof of Theorem 5.6

Proof.

$$\begin{aligned} \mathbb{E}_{y \sim M(x, \textsc{VicLLM}(x))}[u(x, y)] &= \sum_{y \in Y} p_M(y) u(x, y) \\ &= t_\epsilon \sum_{y \in Y} p_M(y) u(x, y) + (1 - t_\epsilon) \mathbb{E}_{y \sim \text{Unif}(S)}[u(x, y)] \end{aligned}$$

Assuming $u(x, y) = 0$ for $y \in S$, we can express the utility ratio:

$$\frac{\mathbb{E}_{y \sim M(x, \textsc{VicLLM}(x))}[u(x, y)]}{\mathbb{E}_{y \sim \textsc{VicLLM}(x)}[u(x, y)]} = \frac{t_\epsilon \sum_{y \in Y} p(y) u(x, y) + (1 - t_\epsilon) \cdot 0}{\sum_{y \in Y} p(y) u(x, y)} = t_\epsilon$$

□

A.4 Proof of Theorem 5.7

Proof. By the chain rule for mutual information:

$$\begin{aligned} I(p_{usr}(\cdot|x^*); (x, Y)) &= I(p_{adv}(\cdot|q), p_{usr}(\cdot|x^*); (x, Y)) - I(p_{adv}(\cdot|q); (x, Y)|p_{usr}(\cdot|x^*)) \\ &= I(p_{adv}(\cdot|q); (x, Y)) + I(p_{usr}(\cdot|x^*); (x, Y)|p_{adv}(\cdot|q)) \\ &\quad - I(p_{adv}(\cdot|q); (x, Y)|p_{usr}(\cdot|x^*)) \\ &\leqslant I(p_{usr}(\cdot|x^*); (x, Y)|p_{adv}(\cdot|q)) + I(p_{adv}(\cdot|q); (x, Y)) \end{aligned}$$

For any malicious query $q \in Q$ and adversary prior $p_{adv} \in \Phi$, we can decompose $I(p_{adv}(\cdot|q); (x, Y))$ into information gain about impermissible A_q and permissible A_q^c conclusions:

$$I(p_{adv}(\cdot|q); (x, Y)) = I_{A_q}(p_{adv}(\cdot|q); (x, Y)) + I_{A_q^c}(p_{adv}(\cdot|q); (x, Y)) \leqslant \epsilon + I_{A_q^c}(p_{adv}(\cdot|q); (x, Y))$$

Using the fact that $I(p_{usr}(\cdot|x^*); (x, Y)|p_{adv}(\cdot|q)) \leqslant \mathcal{H}(p_{usr}(\cdot|x^*)|p_{adv}(\cdot|q))$, we get:

$$I(p_{usr}(\cdot|x^*); (x, Y)) \leqslant \mathcal{H}(p_{usr}(\cdot|x^*)|p_{adv}(\cdot|q)) + I_{A_q^c}(p_{adv}(\cdot|q); (x, Y)) + \epsilon$$

Since this holds for any $q \in Q$ and $p_{adv} \in \Phi$, we can take the infimum:

$$I(p_{usr}(\cdot|x^*); (x, Y)) \leqslant \epsilon + \inf_{\substack{q \in Q \\ p_{adv} \in \Phi}} \Big(\mathcal{H}(p_{usr}(\cdot|x^*)|p_{adv}(\cdot|q)) + I_{A_q^c}(p_{adv}(\cdot|q); (x, Y)) \Big)$$

□

B Decomposition Attack Algorithm Implementation details

B.1 Experimental Details

Dataset Selection and Curation: To evaluate DAs and compare them against jailbreaks under our proposed IIL metric, we chose the WMDP dataset [Li et al., 2024b], focusing on multiple-choice questions in biology and chemistry (WMDP-Bio and WMDP-Chem) as proxies for dangerous knowledge. To assess DAs' ability to extract impermissible information despite filtering mechanisms, we used Llama-Guard-3-8B [Llama Team, 2024] to curate subsets of questions classified as unsafe, resulting in 35 and 64 questions for WMDP-Bio and WMDP-Chem, respectively. This curation ensures the all questions would trigger input filtering mechanisms if asked directly.

Models Used

Model Name	Purpose
Llama-3.1-70B-Instruct [Llama Team, 2024]	Victim model (VICLLM)
Prompt-Guard-86M [Llama Team, 2024]	Jailbreak detection for Input Filtering (M_i)
Llama-Guard-3-8B[Llama Team, 2024]	Input and output filtering mechanism (M_i and M_o)
Mistral-7B-Instruct-v0.3 [Jiang et al., 2023]	Adversary LLM (AdvLLM); not safety alignmed
Llama-3.1-8B-Instruct [Llama Team, 2024]	Adversary LLM (AdvLLM); safety alignmed

Table 3: Models and their purpose in the experiment. All sampling for generation is performed with temperature $t = 0.8$ and nucleus sampling with $p = 0.95$.

Few-shot Examples: Some few-shot examples for subquestion generation E_g and answer aggregation E_a were adapted from [Radhakrishnan et al., 2023] which focused on problem decomposition and aggregation for problem solving. The remaining few shot examples were generated using Claude 3.5 Sonnet [ant, 2024] to match the more technical multiple choice questions in the WMDP dataset.

Hyperparameters: We run our proposed DA with $k = 2$ rounds and $m = 3$ subquestions per round. To reflect more realistic use cases and remove outliers, we only apply the attack to those questions for which the initial answer entropy is greater than 0.5.[1] For each question we run the DA 5 times, returning the IIL measurements for each run as well as the average number of times inputs or outputs were flagged by the input or output filtering models per attack.

[1]This implies that the adversary assigns less than about .97 probability to any one of the choices